

Logical Gates by Gauge Field Formalism of Quantum Error Correction

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Talk plan

- Introduction
- Preliminaries
- Logical gates by gauge field formalism
- Conclusion and outlook

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Why a new framework for logical gates?

- Logical gates are essential for universal quantum computation, but constructing them in QEC codes is **generally non-trivial**
- Existing approaches are **limited to special code families**
 - transversal gates, lattice surgery, code deformation
 - rely on specific geometry or structures
- Gauge field formalism (operator-valued cochain formalism)
 - applicable to **general CSS codes**
 - provides a unified and systematic description of logical gates

What is new?

- Claim (HEP version):
Logical gates are **the gauge-invariant unitaries** acting within the degenerate **ground-state** manifold of CSS codes, characterized by **polynomial** gauge invariants.
- Concrete results:
 - Explicit constructions of gauge-invariant unitaries $(S, H, T, CZ, C^{m-1}Z)$
 - Logical action depends only on homology class
(= gauge equivalence of Wilson-loop representatives)
 - Direct microscopic (physical-gate) realizations
- Take-home message:
Logical gates are determined **algebraically by gauge constraints**, independent of geometry.

What we achieve

- General framework
 - Logical gates for general CSS codes
 - Gauge field (operator-valued cochain) formalism
- Constructed logical gates
 - S, H, T, CZ, multi-controlled-Z
 - No geometric or product structure assumed
- Key properties
 - exponential of polynomials of gauge fields
 - explicit physical decompositions
 - depend only on (co)homology classes
- Scope
 - No fault tolerance or depth optimality assumed
 - Focus: algebraic existence and structure

Contribution overview

- Summary of results:
 - Unified construction for general CSS codes (no geometry assumed)
- We explain S gate in this talk
as an illustration of how gauge-field expressions encode logical gates

Logical Gate	Physical Gate Decomposition	Gauge Field Expression
$\overline{Z}(\gamma)$	$\prod_k Z_{i_k}$	$e^{i\pi a(\gamma)}$
$\overline{X}(\tilde{\gamma})$	$\prod_k X_{\tilde{i}_k}$	$e^{i\pi b(\tilde{\gamma})}$
$\overline{S}(\gamma)$	$\prod_k S_{i_k} \prod_{k_1 < k_2} CZ_{i_{k_1}, i_{k_2}}$	$e^{i\frac{\pi}{2}a(\gamma)^2}$
$\overline{H}(\gamma)$	$e^{-i\frac{\pi}{4}} \overline{S}(\gamma) \cdot \prod_k H_{\tilde{i}_k} \cdot \overline{S}(\tilde{\gamma}) \cdot \prod_k H_{\tilde{i}_k} \cdot \overline{S}(\gamma)$	$e^{-i\frac{\pi}{4}} e^{i\frac{\pi}{2}a(\gamma)^2} e^{i\frac{\pi}{2}b(\tilde{\gamma})^2} e^{i\frac{\pi}{2}a(\gamma)^2}$
$\overline{CZ}(\gamma_1, \gamma_2)$	$\prod_{k_1, k_2} CZ_{i_{k_1}^1, i_{k_2}^2}$	$e^{i\pi a(\gamma_1)a(\gamma_2)}$
$\overline{C^{m-1}Z}(\gamma_1, \dots, \gamma_m)$	$\prod_{k_1, \dots, k_m} C^{m-1}Z_{i_{k_1}^1, \dots, i_{k_m}^m}$	$e^{i\pi a(\gamma_1) \dots a(\gamma_m)}$
$\overline{T}(\gamma)$	$\prod_k T_{i_k} \prod_{k_1 < k_2} CS_{i_{k_1}, i_{k_2}}^\dagger \prod_{k_1 < k_2 < k_3} CCZ_{i_{k_1}, i_{k_2}, i_{k_3}}$	$e^{i\frac{\pi}{4}(2a(\gamma)^3 - 3a(\gamma)^2 + 2a(\gamma))}$

One slide for HEP audience

- CSS code = Z_2 lattice gauge theory on two-dimensional hypergraphs without dynamics
- Stabilizers = Plaquette terms/Gauss law constraints
- Logical Pauli operators = Wilson loops (homology classes)
- Logical gates = gauge-invariant unitaries acting on ground-state manifold
- This talk: classify gauge-invariant unitaries via polynomial of loop integrals in gauge fields

(No dynamics, no path integral, no continuum limit)

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CSS code (Review)

- CSS code: commutative X -type or Z -type stabilizer generators
- Code space: simultaneous eigenstates with $+1$ eigenvalue

$$S_i^X |\psi\rangle = S_i^Z |\psi\rangle = +|\psi\rangle$$

$\Rightarrow$ encodes logical information with redundancy

- Parity check matrix H_X, H_Z ($H_X H_Z^T = 0$) gives linear constraints

$$S_i^X =: \prod_j (X_j)^{(H_X)_{ij}}, \quad S_i^Z = \prod_j (Z_j)^{(H_Z)_{ij}}$$

- Logical Pauli operators
 - constructed as products as Pauli X/Z operators
 - classified by homology ($\ker H_X / \text{Im } H_Z^T$)

Chain complex (Review)

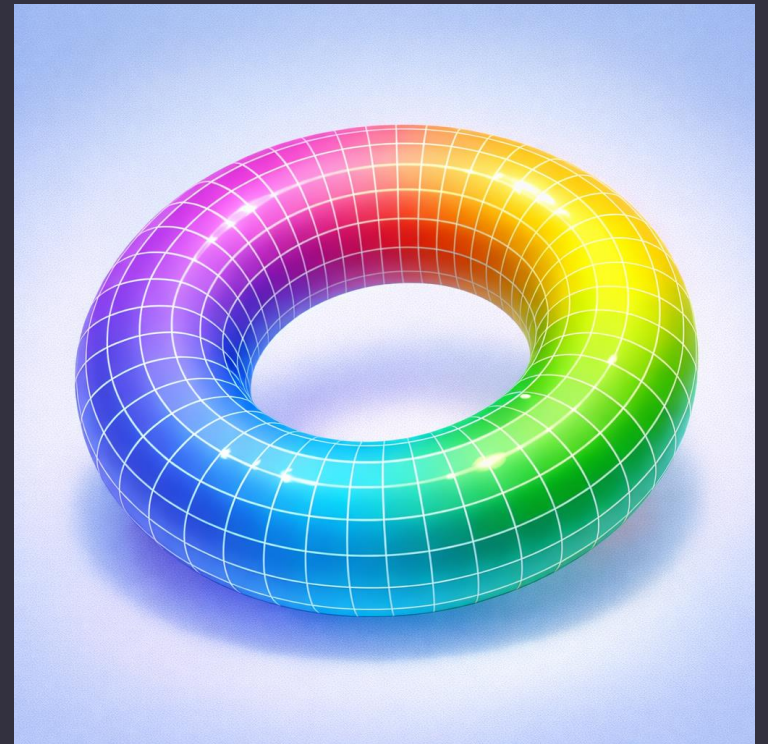
- A method to study **geometric structure algebraically**
- Chain complex : vector spaces $\{C_n\}$ with linear maps $\{\partial_n\}$

$$\cdots \xrightarrow{\partial_{n+2}} C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}} \cdots$$

- Boundary operator ∂_n satisfies $\partial_n \partial_{n+1} = 0$
(“Boundaries of boundaries” vanish)
- Homology groups H_n detect global topological degrees of freedom:

$$H_n := \ker \partial_n / \operatorname{im} \partial_{n+1}$$

(represents loops that are not boundaries)



CSS-homology correspondence (Review)

- Any CSS code corresponds to a length-2 chain complex:

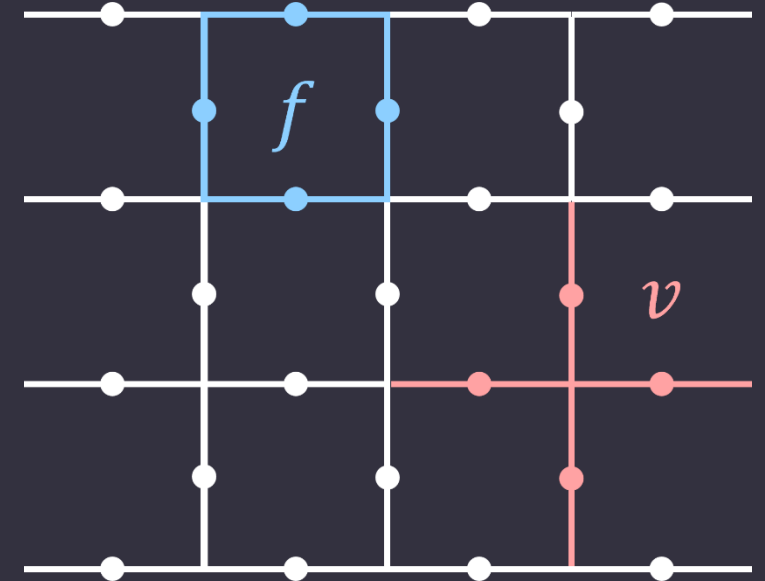
$$C_2 \xrightarrow{\partial_2 = H_Z^T} C_1 \xrightarrow{\partial_1 = H_X} C_0$$

- Dictionary:

– Parity check matrix	$\leftrightarrow$ Boundary operator ($\partial_2 = H_Z^T, \partial_1 = H_X$)
– Physical qubit	$\leftrightarrow$ Edge ($C_1 = \mathbb{F}_2^{n_x}$)
– X Stabilizer	$\leftrightarrow$ Vertex ($C_0 = \mathbb{F}_2^{n_x}$)
– Z stabilizer	$\leftrightarrow$ Face ($C_2 = \mathbb{F}_2^{n_z}$)
– Logical Pauli operator	$\leftrightarrow$ (Co)homology class

$$\bar{Z}(\gamma) : \gamma \in H_1 = \ker H_X / \text{im } H_Z^T$$

- Notice: Euclidean embeddings may NOT exist
 $\rightarrow$ Hypergraphs in general



(e.g.) Toric code

Gauge field in QEC: analogy (Review)

- Gauge transformation $\leftrightarrow$ Local transformation generated by X stabilizers
(Gauss law constraint/star term)

- What is or is not a gauge field here:

Is – Z_2 lattice gauge field
– lives on edges
– Wilson loop = logical Pauli
(homology class)

Is not – dynamical field
– path integral
– effective field theory
– continuum gauge theory

- Toric-code-type Hamiltonian

$$H = - \sum_f S^Z(f) - \sum_v S^X(v)$$

$\Rightarrow$ The ground-state manifold gives the code space.
Used only as a kinematic anchor; no dynamics is discussed.

Gauge field formalism (Review)

- Gauge field $a(\gamma), b(\gamma)$:
operator-valued 1-cochains on qubits (linear maps on C_1)

$$a(\gamma_1 + \gamma_2) = a(\gamma_1) + a(\gamma_2)$$

- For the j -th qubit e_j ,

$$a(e_j) = z_j, \quad b(e_j) = x_j$$

where $z := P^0 = |0\rangle\langle 0|$, $x := P^- = |-\rangle\langle -|$

- Pauli operators are exponentials of gauge fields:

$$Z = \exp(i\pi z) = I - 2z, \quad X = \exp(i\pi x) = I - 2x$$

$\Rightarrow a(\gamma)/b(\gamma)$ is a “logarithm” of Pauli Z/X strings

Gauge field formalism (Review)

- Gauge field formalism = CSS code written in operator-valued cochains
- Transversal Pauli operators are expressed as exponentials of gauge fields

$$\bar{Z}(\gamma) = \exp(i\pi a(\gamma)) = \exp\left(i\pi \sum_i \gamma^i a(e_i)\right) = \prod_i (Z_i)^{\gamma^i}$$

$$\bar{X}(\gamma) = \exp(i\pi b(\gamma)) = \exp\left(i\pi \sum_i \gamma^i b(e_i)\right) = \prod_i (X_i)^{\gamma^i}$$

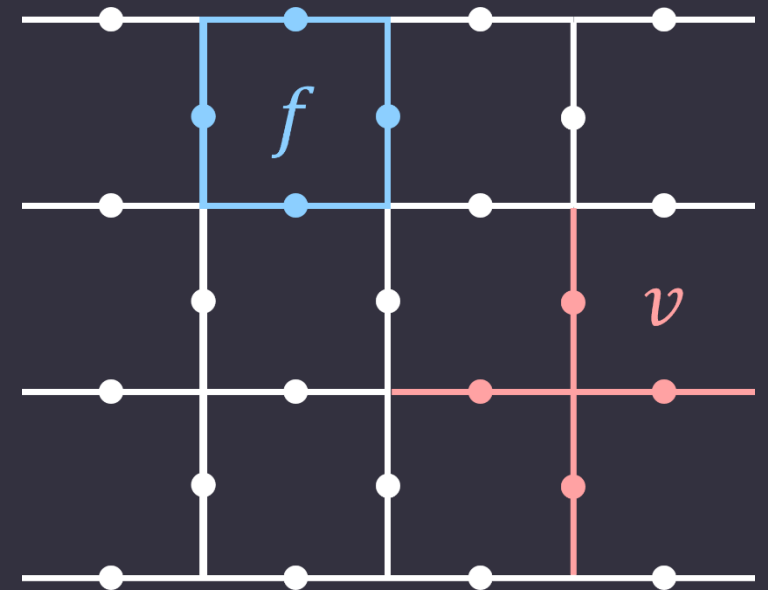
$\gamma \in \mathcal{C}_1$: subset of physical qubits (1-chain)

$a(\gamma), b(\gamma)$: operator-valued electric/magnetic gauge field

- Stabilizers are expressed as

$$S^Z(f) = \bar{Z}(\partial_2 f), \quad S^X(v) = \bar{X}(\partial_1^T v)$$

$\Rightarrow$ X/Z stabilizers correspond to exact 1-chains (boundaries/coboundaries)



Gauge field formalism (Review)

- Commutation relations encoded in **intersections**

$$\bar{X}(\gamma_2)a(\gamma_1)\bar{X}(\gamma_2) = a(\gamma_1) - 2a(\gamma_1 \cap \gamma_2) + \langle \gamma_1, \gamma_2 \rangle$$

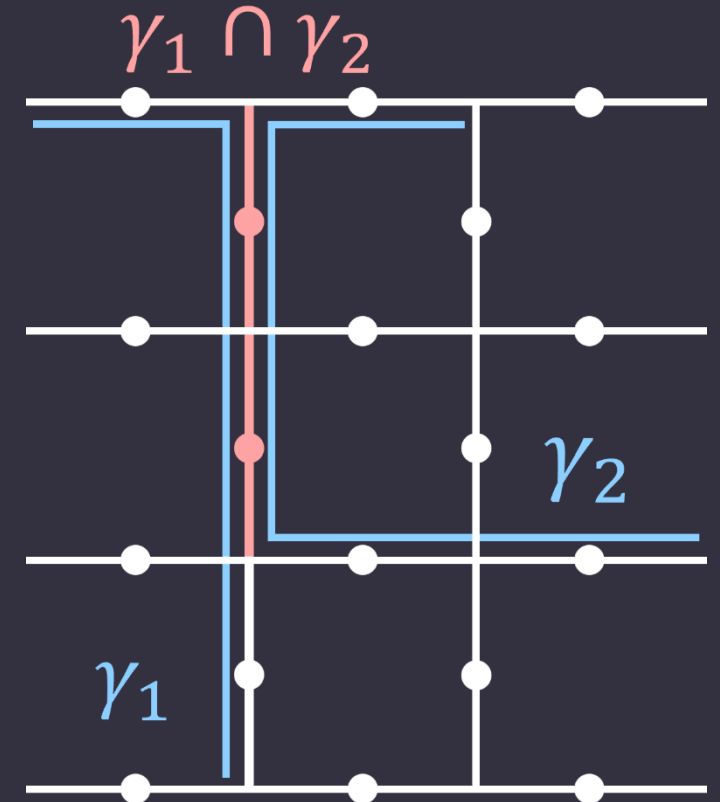
$$\bar{Z}(\gamma_2)b(\gamma_1)\bar{Z}(\gamma_2) = b(\gamma_1) - 2b(\gamma_1 \cap \gamma_2) + \langle \gamma_1, \gamma_2 \rangle$$

$\Rightarrow$ X/Z strings **locally** modify gauge fields on the intersection ($\gamma_1 \cap \gamma_2 := \sum_{j=1}^n \gamma_1^j \gamma_2^j e_j$)

- Commutation relation between X and Z:

$$\bar{X}(\gamma_2)\bar{Z}(\gamma_1)\bar{X}(\gamma_2) = \exp(i\pi\langle \gamma_1, \gamma_2 \rangle) \bar{Z}(\gamma_1)$$

$\Rightarrow$ Pauli commutation is completely determined by the **intersection number** ($\langle \gamma_1, \gamma_2 \rangle := \sum_j \gamma_1^j \gamma_2^j$)



Stabilizers as gauge transformations

- Stabilizer conjugation = Z_2 gauge transformation of gauge fields
- Correspondence
 - 0-cochain $\tilde{v}$ $\leftrightarrow$ gauge parameter
 - coboundary $d\tilde{v}$ $\leftrightarrow$ gauge transformation
 - stabilizer conjugation $\leftrightarrow$ gauge redundancy
- 0-cochain $\tilde{v}(v)$ and its exterior derivative $d\tilde{v}(\gamma)$:

$$\tilde{v}(v') := \langle v, v' \rangle, \quad d\tilde{v}(\gamma) := \tilde{v}(\partial_1 \gamma) = \langle v, \partial_1 \gamma \rangle$$

- Conjugation by the stabilizers gives

$$S^X(v) a(\gamma) S^X(v) = a(\gamma) + d\tilde{v}(\gamma) - 2a(\gamma \cap \partial_1^T v)$$

$\Rightarrow Z_2$ gauge transformation
(the last term is physically invisible modulo 2)

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 - S gate (in detail)
 - Other gates (only the structure and key ideas)
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Scope and guiding question

- Up to here: kinematics of CSS codes as a Z_2 lattice gauge theory.
- From now on: which gauge-invariant unitaries act nontrivially on the ground-state sector?
- Scope of this talk
 - Purely kinematical (ground-state sector only)
 - Discrete Z_2 gauge fields (operator-valued cochains)
 - No dynamics, no EFT, no continuum limit
 - Focus: algebraic structure of logical gates
(gauge invariant-unitaries on the code space)

Main question (HEP version)

Which unitary operators preserve the ground-state sector,
and how much structure do gauge constraints impose?

- We do not give a complete classification.
Instead, we identify a large, structured family:
unitaries generated by exponentials of polynomial gauge invariants
- Polynomial degree $\leftrightarrow$ level of Clifford hierarchy
(level- k unitaries map Pauli operators to
level- $(k-1)$ operators under conjugation)
- Dictionary (QEC $\leftrightarrow$ HEP):
 - Code space $\leftrightarrow$ ground states
(= degenerate ground-state manifold)
 - Logical gates $\leftrightarrow$ gauge-invariant unitary operators
(acting within the ground-state sector)

Three ideas to construct logical gates

1. Any function of logical $\bar{X}, \bar{Z}$ is logical
(commutes with all stabilizers)

$$[g(\bar{X}, \bar{Z}), S^Z(f)] = [g(\bar{X}, \bar{Z}), S^X(v)] = 0$$

2. Replacement principle

Physical gate expression $\Rightarrow$ Logical expression

$$\mathcal{O} = g(X, Z) \Rightarrow \bar{\mathcal{O}} = g(\bar{X}, \bar{Z})$$

3. Projector decomposition

Pauli operator $\rightarrow$ polynomials $\rightarrow$ exponentials

$$X^c = I - 2cx, \quad Z^c = I - 2cz \quad (c = 0, 1)$$

$\Rightarrow$ enables gauge-field expression

Why gauge fields are useful for logical gates

1. Logical action depends **only on the homology class**
(independent of representative choice)

$$\mathcal{O}(\gamma) \sim \mathcal{O}(h(\gamma))$$

⇒ Logical gates are not operators on physical qubits,
but on **equivalence classes of loops** (logical qubits).
Gauge fields are the **minimal** variables that remember this equivalence.

2. **Clifford and non-Clifford** gates are treated on
the same footing as exponentials of gauge-field polynomials
⇒ Clifford hierarchy = **polynomial degree** of gauge fields

Examples

- Clifford : $\exp(i\pi a(\gamma)^2 / 2), \exp(i\pi a(\gamma_1)a(\gamma_2))$
- Non-Clifford : $\exp(i\pi a(\gamma)^3 / 2), \exp(i\pi a(\gamma_1)a(\gamma_2)a(\gamma_3))$

Polynomial gauge invariants

- Claim:
Logical gates are generated by exponential of gauge-invariant polynomials,
with polynomial degree = Clifford level
- Results:

degree	operator	example	interpretation
1	linear	Pauli/Wilson loop	l-form symmetry generator
2	quadratic	S, CZ, H	bilinear gauge invariant
3	cubic	T	cubic gauge invariant
m	m -linear	$C^{m-1}Z$	higher polynomial invariant

- Notice: polynomials are taken after loop integration
 $a(\gamma)^m \leftrightarrow (\oint_{\gamma} a)^m$, not $\oint_{\gamma} a^m$ (discrete Z_2 gauge fields/projector-valued)

S gate

- Definition

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \exp\left(\frac{i\pi}{2} \frac{I - Z}{2}\right)$$

⇒ diagonal gate = exponential of polynomial of Pauli Z

- Commutation relation

$$XSX = iZS$$

⇒ determines **action on logical X**

- Why S gate?

- diagonal gate → electric gauge field $a(\gamma)$ only
- minimal nontrivial polynomial : $\exp(i\pi a(\gamma)^2/2)$
- prototype for logical gates : homology dependence + CZ structure

- Take-home message:

Logical S = quadratic gauge field ⇒ Clifford & homology-protected

Logical S gate

- Replacement principle $\Rightarrow$ candidate of logical S gate

$$\bar{S}(\gamma) = \exp\left(\frac{i\pi I - \bar{Z}(\gamma)}{2}\right) \quad (\text{replacement: } Z \rightarrow \bar{Z}(\gamma))$$

- Check: commutation relation with logical X

$$\begin{aligned} \bar{X}(\tilde{\gamma})\bar{S}(\gamma)\bar{X}(\tilde{\gamma}) &= \exp\left(\frac{i\pi I - \bar{X}(\tilde{\gamma})\bar{Z}(\gamma)\bar{X}(\tilde{\gamma})}{2}\right) && (\text{definition}) \\ &= \exp\left(\frac{i\pi I + \bar{Z}(\gamma)}{2}\right) && (\text{since } \bar{X}(\tilde{\gamma})\bar{Z}(\gamma)\bar{X}(\tilde{\gamma}) = -\bar{Z}(\gamma)) \\ &= \exp\left(\frac{i\pi}{2}\bar{Z}(\gamma)\right)\exp\left(\frac{i\pi I - \bar{Z}(\gamma)}{2}\right) && (\text{rearranging commuting terms}) \\ &= i\bar{Z}(\gamma)\bar{S}(\gamma) \end{aligned}$$

$\Rightarrow$ Correct Clifford action on the logical Pauli algebra

From logical Z to physical gates

(The most technical part)

Example: $\bar{Z} = Z_1 Z_2 Z_3$

- Goal: $\bar{S} = \exp(i\pi/2 \cdot (I - \bar{Z})/2) \rightarrow$ physical S and CZ gates

Claim (spoiler!): $\bar{S} = S_1 S_2 S_3 \cdot CZ_{1,2} CZ_{1,3} CZ_{2,3}$

- Strategy:

1. Rewrite $\bar{Z}$ using projectors z_i

2. Plug into the exponential

3. Interpret each exponential term as a physical gate

- Why projectors z_i ?

$\rightarrow \underline{z_i^2 = z_i}$, eigenvalues = 0,1 (integer spectrum)

- Useful relation

- $Z_i = I - 2z_i$

- $(I - Z_1 Z_2 Z_3)/2 = z_1 + z_2 + z_3 - 2(z_1 z_2 + z_1 z_3 + z_2 z_3) + 4z_1 z_2 z_3$

(This is a discrete analogue of expanding a gauge-invariant operator in integer-valued fields.)

From logical Z to physical gates

(The most technical part)

Example: $\bar{Z} = Z_1 Z_2 Z_3$

- Key: $\bar{S} = \exp(i\pi/2 \cdot (I - \bar{Z})/2)$ factorizes into S and CZ
- Using the relation from the previous slide:

$$\begin{aligned}(I - Z_1 Z_2 Z_3)/2 &= z_1 + z_2 + z_3 - 2(z_1 z_2 + z_1 z_3 + z_2 z_3) + 4z_1 z_2 z_3 \\ \Rightarrow \bar{S} &= \exp\left(\frac{i\pi}{2}(z_1 + z_2 + z_3)\right) \exp(-i\pi(z_1 z_2 + z_1 z_3 + z_2 z_3)) \exp(2\pi i z_1 z_2 z_3) \\ &= \left(\prod_{i=1}^3 \exp(i\pi z_i/2)\right) \left(\prod_{i<j} \exp(-i\pi z_i z_j)\right) \exp(2\pi i \Delta)\end{aligned}$$

- Dictionary: $\exp(i\pi z_i/2) = S_i$, $\exp(-i\pi z_i z_j) = CZ_{i,j}$
- Higher-order terms: $\Delta = z_1 z_2 z_3$ (integer eigenvalues) $\Rightarrow \exp(2\pi i \Delta) = 1$

Decomposition to physical gates

- Result: logical S decomposes into physical S and CZ on $\text{supp}(\gamma)$

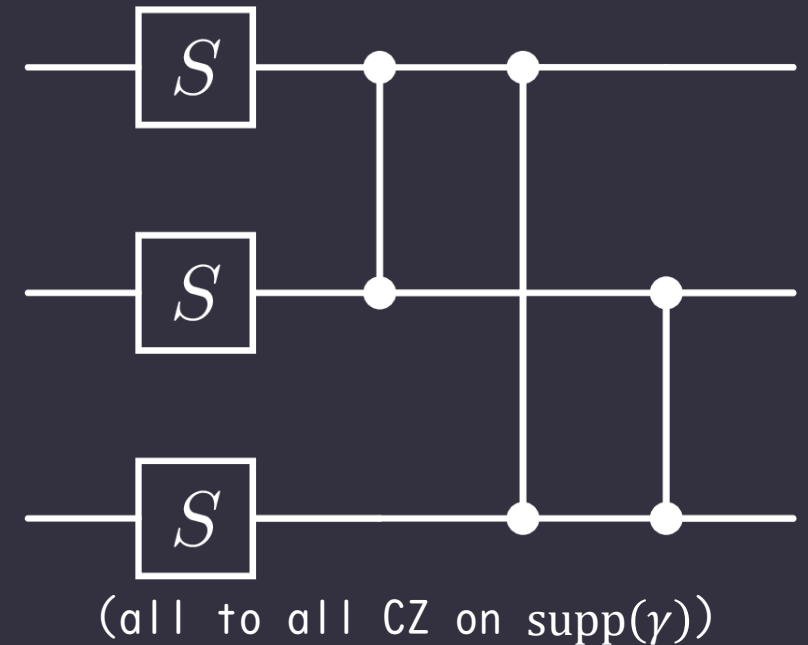
$$\bar{S}(\gamma) = \prod_i (S_i)^{\gamma^i} \prod_{i < j} (CZ_{ij})^{\gamma^i \gamma^j}$$

- Since $\gamma^i \in \{0,1\}$, only qubits in $\text{supp}(\gamma)$ contribute:

$$\bar{S}(\gamma) = \prod_{i \in \text{supp}(\gamma)} S_i \prod_{i < j} CZ_{ij}$$

- Definition of support of γ :

$$\text{supp}(\gamma) := \{j \in \{1, \dots, n\} \mid \gamma^j = 1\}$$



Gauge field expression of logical S

- Rewriting the projector expansion in terms of gauge fields

$$\sum_i \gamma^i z_i = a(\gamma) \quad (\text{linear term})$$

$$\sum_{i < j} \gamma^i \gamma^j z_i z_j = \frac{1}{2} a(\gamma)^2 - \frac{1}{2} a(\gamma) \quad (\text{quadratic term})$$

- Result: compact gauge-field form

$$\begin{aligned} \bar{S}(\gamma) &= \exp\left(\frac{i\pi}{2} a(\gamma)\right) \exp\left(\frac{i\pi}{2} (a(\gamma)^2 - a(\gamma))\right) \\ &= \exp\left(\frac{i\pi}{2} a(\gamma)^2\right) \end{aligned}$$

→ Quadratic polynomial $\Rightarrow$ Clifford gate

Homology dependence of logical S gate

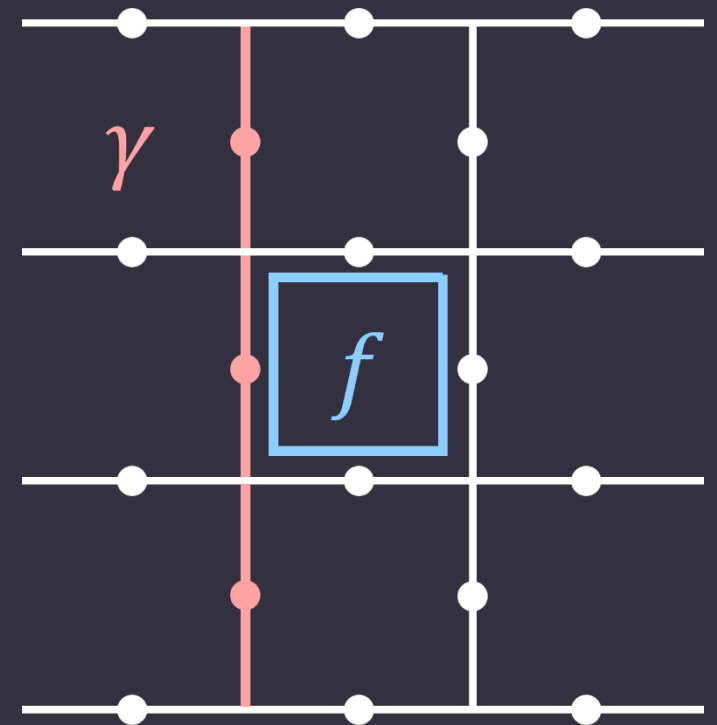
- Logical S gate depends **only on the homology class** of γ :

$$\gamma_1 = \gamma_2 + \partial_2 f \Rightarrow \bar{S}(\gamma_1) \sim \bar{S}(\gamma_2)$$

($\sim$: identical action on the code space)

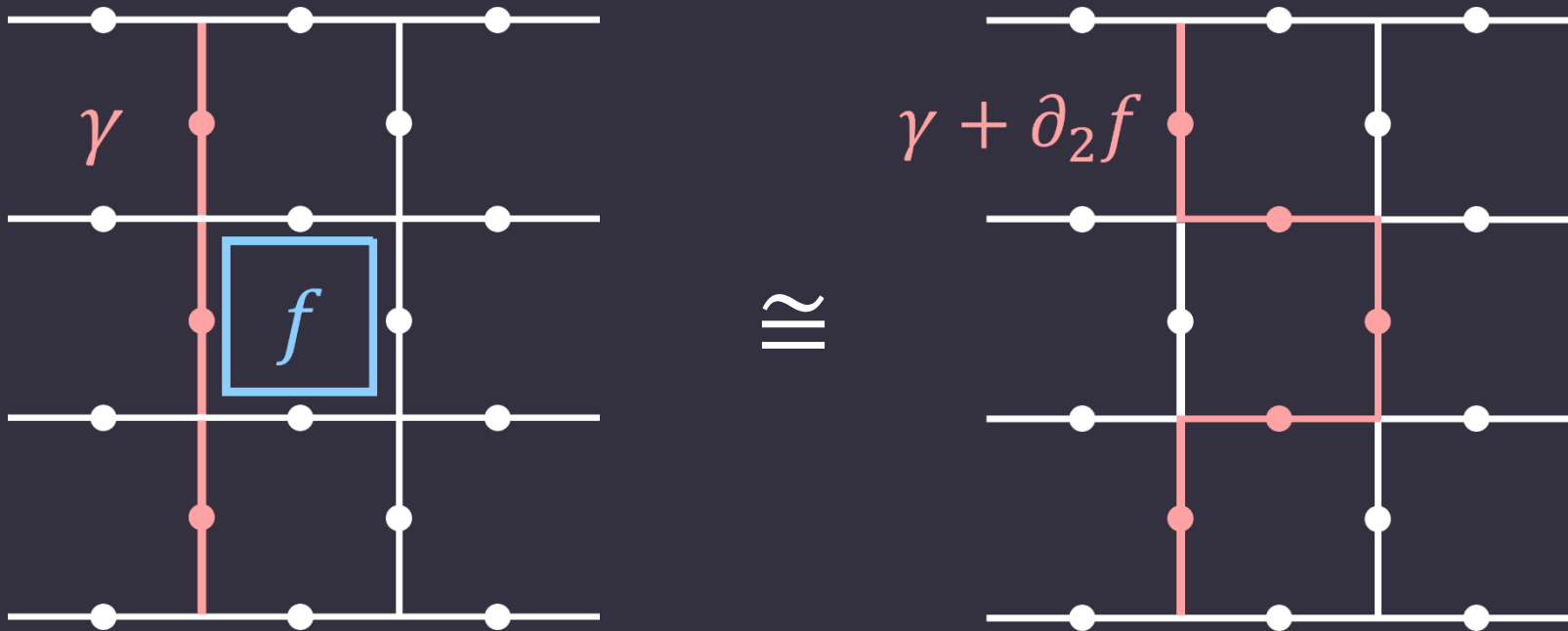
$\Rightarrow \bar{S}(\gamma) \sim \bar{S}(h(\gamma))$: **well-defined on the code space**

- Deforming γ by stabilizers (boundaries) changes the physical circuit, but not the logical action
- Intuition:
Boundary terms generate (controlled) stabilizers
 $\Rightarrow$ trivial action on the code space.
(These terms generate controlled interactions, but remain gauge-trivial on the code space.)



Homology = topological equivalence

- Deformation by boundary (= stabilizer multiplication)
→ gauge-equivalent operator (topological deformation)
- Logical action on the code space should be unchanged
- Equivalent to deformation of Wilson loop



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Hadamard: alternating quadratic forms

- Non-diagonal Clifford gate (mixes X and Z)

$$H := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad HZH = X, \quad HXH = Z$$

- Euler angle definition
→ written only with diagonal Z/X exponentials

$$H = \exp\left(-\frac{i\pi}{4}\right) \exp\left(\frac{i\pi I - Z}{2}\right) \exp\left(\frac{i\pi I - X}{2}\right) \exp\left(\frac{i\pi I - Z}{2}\right)$$

- Logical gate ($\tilde{\gamma}$: dual logical chain $\langle \gamma, \tilde{\gamma} \rangle = 1$)

$$\bar{H}(\gamma) = \exp\left(-\frac{i\pi}{4}\right) \exp\left(\frac{i\pi I - \bar{Z}(\gamma)}{2}\right) \exp\left(\frac{i\pi I - \bar{X}(\tilde{\gamma})}{2}\right) \exp\left(\frac{i\pi I - \bar{Z}(\gamma)}{2}\right)$$

can be written as a product of $\exp(i\pi a(\gamma)^2/2)$ and $\exp(i\pi b(\gamma)^2/2)$

Hadamard: alternating quadratic forms

- Logical Hadamard = **alternating electric/magnetic** quadratic gauge fields
($\Rightarrow$ non-diagonal Clifford, but still homology-protected)

- Physical gate decomposition

$$\bar{H}(\gamma) = e^{-\frac{i\pi}{4}} \bar{S}(\gamma) H(\tilde{\gamma}) \bar{S}(\tilde{\gamma}) H(\tilde{\gamma}) \bar{S}(\gamma)$$

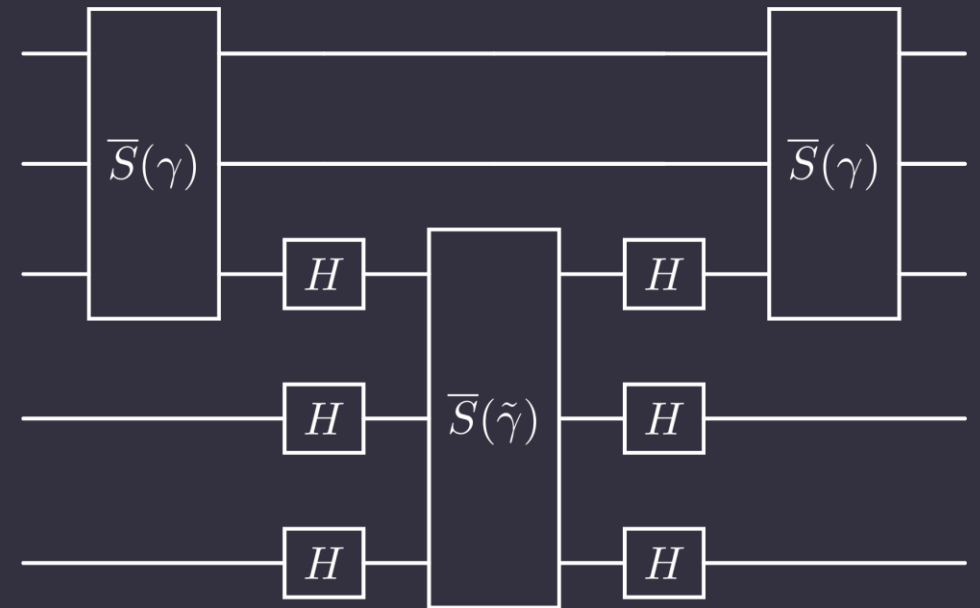
$\rightarrow$ many layers of S, H, CZ (X-Z mixing)

- Gauge field expression

$$\bar{H}(\gamma) = e^{-\frac{i\pi}{4}} e^{i\pi a(\gamma)^2/2} e^{i\pi b(\tilde{\gamma})^2/2} e^{i\pi a(\gamma)^2/2}$$

$$(\gamma \in H_1 \text{ (logical } Z), \tilde{\gamma} \in H^1 \text{ (logical } X), \langle \gamma, \tilde{\gamma} \rangle = 1)$$

- Homology dependence: $\bar{H}(\gamma_1) \sim \bar{H}(\gamma_2)$
boundaries only generate stabilizers
 $\Rightarrow$ logical action is homology invariant



(not depth-optimal, shown for existence)

Controlled-Z: bilinear gauge field

$$CZ_{i,j} = I_j P_i^1 + Z_j P_i^0 = \exp\left(i\pi \frac{I - Z_i}{2} \frac{I - Z_j}{2}\right) \quad (\text{projector form})$$

- Logical gate by replacement

$$\overline{CZ}(\gamma_1, \gamma_2) = \exp\left(i\pi \frac{I - \bar{Z}(\gamma_1)}{2} \frac{I - \bar{Z}(\gamma_2)}{2}\right)$$

- Physical decomposition (**round-robin construction**)

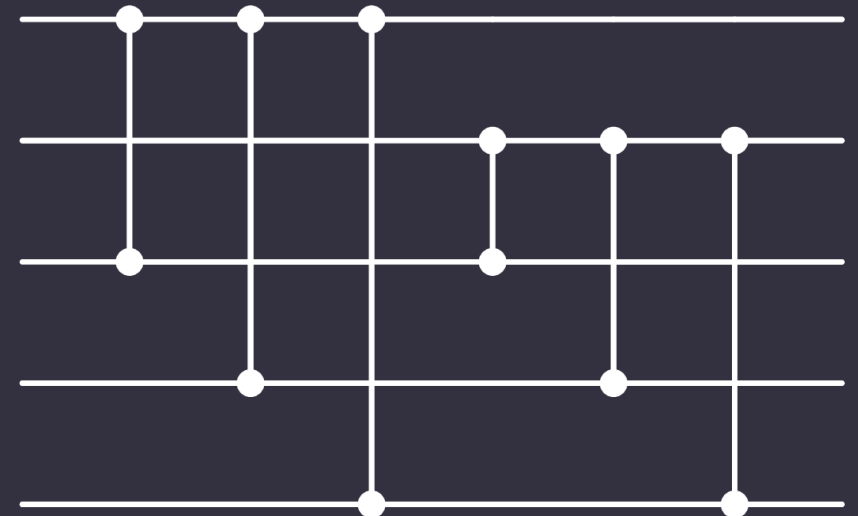
$$\overline{CZ}(\gamma_1, \gamma_2) = \prod_{i \in \text{supp}(\gamma_1)} \prod_{j \in \text{supp}(\gamma_2)} CZ_{i,j}$$

Gauge field expression

$$\overline{CZ}(\gamma_1, \gamma_2) = \exp(i\pi a(\gamma_1) a(\gamma_2))$$

$\Rightarrow$ **bilinear form** of gauge fields

[Joder, Takagi, Chuang, 2016]



CZ and multi-controlled-Z: multi-linear

- Logical CZ depends only on homology classes

$$\overline{CZ}(\gamma_1, \gamma_2) \sim \overline{CZ}(\gamma'_1, \gamma'_2), \quad (\gamma_i - \gamma'_i \in \text{im } \partial_2)$$

- Physical decomposition (round-robin)

$$\overline{C^{m-1}Z}(\gamma_1, \dots, \gamma_m) = \prod_{i_1 \in \text{supp}(\gamma_1)} \dots \prod_{i_m \in \text{supp}(\gamma_m)} C^{m-1}Z_{i_1, \dots, i_m}$$

→ apply a **physical $C^{m-1}Z$** for **every choice of qubits**
in the supports of logical operators

- Gauge field expression

[Breuckmann, et.al, 2024]

$$\overline{C^{m-1}Z}(\gamma_1, \dots, \gamma_m) = \exp(i\pi a(\gamma_1) \cdots a(\gamma_m))$$

⇒ **multi-linear polynomial** of gauge fields
(degree $m \Rightarrow$ Clifford hierarchy level m)

T: physical gate decomposition

- Non-Clifford diagonal gate

$$T = \exp\left(\frac{i\pi}{4} \frac{I - Z}{2}\right)$$

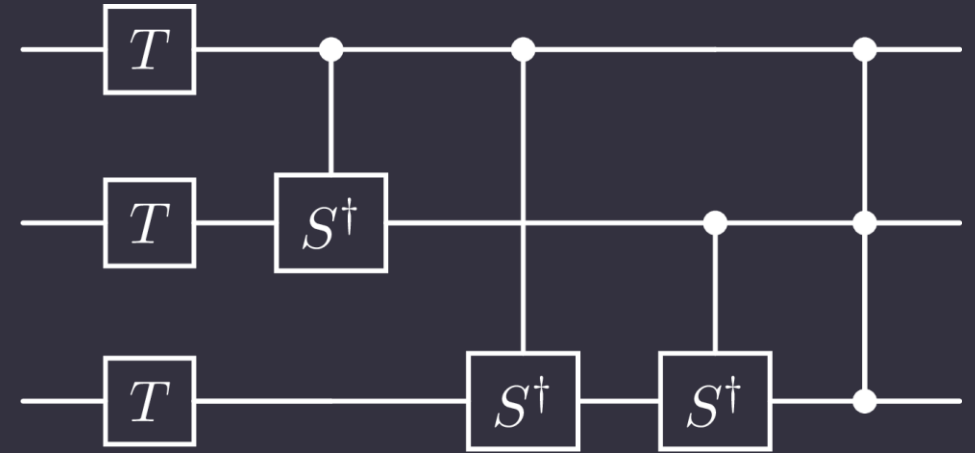
- Physical gate decomposition

$$\bar{T}(\gamma) = \left(\prod_{i \in \text{supp}(\gamma)} T_i \right) \left(\prod_{i < j} C S_{i,j}^\dagger \right) \left(\prod_{i < j < k} C C Z_{i,j,k} \right)$$

1Q gate

2Q gate

3Q gate



(Example for $|\text{supp}(\gamma)| = 3$)

- Key point:

Non-Clifford diagonal gates require **higher-order controlled-Z gates**

T: cubic gauge field expression

$$\bar{T}(\gamma) = \exp\left(i\pi\left(\frac{1}{2}a(\gamma)^3 - \frac{3}{4}a(\gamma)^2 + \frac{1}{2}a(\gamma)\right)\right)$$

⇒ cubic polynomial in gauge field

- Logical T gate depends only on the homology class of γ :

$$\bar{T}(\gamma) \sim \bar{T}(\gamma + \partial_2 f)$$

- Diagonal logical gates are classified by gauge-field polynomials
(Polynomial degree $\leftrightarrow$ Clifford hierarchy level)

Talk plan

- Introduction
- Preliminaries
- Logical gates by gauge field formalism
- Conclusion and outlook

Our contribution

- Systematic construction of logical gates for **general CSS codes**
- Gauge-field formalism (operator-valued cochain formalism)
 - Logical gates as **exponential of polynomials of gauge fields**
- Logical action depends only on **(co)homology classes**
 - independent of representative choice
- Scope:
 - **No fault-tolerance or depth-optimality** assumed
 - Focus on algebraic existence and structure

Outlook

- Gauge fields provide a common language for logical gates, independent of geometry
- 1. For QEC: **beyond kinematics**
 - Measurement-based schemes
(quantum channels as dynamical processes)
 - Longer (= higher dim.) chain complexes
(measurement error, single-shot QEC)
- 2. For HEP: **beyond Z_2 gauge structure** in CSS codes
 - Subsystem codes
 - Non-CSS stabilizer codes
 - Non-Abelian finite group
 - Higher-form symmetries



(preprint and extended details)

Backup

Applicability of our construction

- Applicable
 - Diagonal gates

$$\mathcal{O}_k(i_1, \dots, i_m) := \exp \left(\frac{i\pi}{2^k} \frac{I - Z_{i_1}}{2} \dots \frac{I - Z_{i_m}}{2} \right)$$

- Their variants with X/Y instead of Z
 - Alternating layer of them
- Not applicable
 - Non-diagonal 2- or more qubits gate with overlapping supports
e.g.) CNOT

$$\overline{CNOT}(\gamma_1, \gamma_2) = \exp \left(i\pi \frac{I - \bar{Z}(\gamma_1)}{2} \frac{I - \bar{X}(\gamma_2)}{2} \right)$$

→ If $\gamma_1 \cap \gamma_2 \neq 0$, gauge fields do not commute
(Another option: utilizing $CNOT_{i,j} = H_j C Z_{i,j} H_j \rightarrow$ deep circuit)

Polynomial of gauge fields

- Polynomial of gauge fields in terms of projectors

$$a(\gamma)^m = \sum_{r=1}^m r! S(m, r) \sum_{i_1 < \dots < i_r} \gamma^{i_1} \dots \gamma^{i_r} z_{i_1} \dots z_{i_r}$$

($S(m, r)$: the Stirling number of the second kind)

- Polynomial of projectors in terms of gauge fields

$$\sum_{i_1 < \dots < i_r} \gamma^{i_1} \dots \gamma^{i_r} z_{i_1} \dots z_{i_r} = \frac{1}{r!} \sum_{m=1}^r (-1)^{m+r} s(r, m) a(\gamma)^m$$

($s(r, m)$: the Stirling number of the first kind)

Homology dependence of S gate

- We show $\bar{S}(\gamma) \sim \bar{S}(\gamma + \partial_2 f) = \bar{S}(\gamma) \cdot \exp(i\pi a(\gamma)a(\partial_2 f)) \cdot \bar{S}(\partial_2 f)$
- The second and third terms are logically identity operator (i.e. $\mathcal{O}|\psi\rangle = |\psi\rangle$ for any logical state $|\psi\rangle$)
 - The second term is a product of controlled-Z stabilizer gate:

$$\exp(i\pi a(\gamma)a(\partial_2 f)) = \prod_{i \in \text{supp}(\gamma)} (P_i^+ + P_i^- S^Z(\partial_2 f))$$

- The third term is expressed as

$$\bar{S}(\partial_2 f) = \exp\left(\frac{i\pi I - S^Z(\partial_2 f)}{2}\right)$$

$\Rightarrow$ Identity operators on the code space since $S^Z(\partial_2 f) \sim I$

- Therefore, $\bar{S}(\gamma + \partial_2 f) = \bar{S}(\gamma) \cdot I \cdot I \sim \bar{S}(\gamma)$

Homology dependence in general

- The homology dependence holds for any logical operators constructed by the replacement principle
- Example: diagonal logical gate with a polynomial expansion

$$\bar{\mathcal{O}}(\gamma) = g(\bar{Z}(\gamma)) = \sum_n c_n \bar{Z}(\gamma)^n$$

- Replacing $\gamma \rightarrow \gamma + \partial_2 f$ gives $\bar{\mathcal{O}}(\gamma + \partial_2 f) = \sum_n c_n (\bar{Z}(\gamma) S^Z(f))^n$
- Since $S^Z(f) \sim I$, we have

$$\bar{\mathcal{O}}(\gamma + \partial_2 f) \sim \sum_n c_n (\bar{Z}(\gamma) \cdot I)^n = \bar{\mathcal{O}}(\gamma)$$

- This discussion can be extended straightforwardly to the case of non-diagonal gates and multiple logical qubits $\bar{\mathcal{O}} = g(\bar{Z}_1, \dots, \bar{Z}_k, \bar{X}_1, \dots, \bar{X}_k)$

Multi-controlled-Z gate

- Definition

$$C^{m-1}Z_{i_1, \dots, i_m} = \exp\left(i\pi \frac{I - Z_{i_1}}{2} \cdots \frac{I - Z_{i_m}}{2}\right)$$

- Gauge field expression

$$\overline{C^{m-1}Z}(\gamma_1 \cdots \gamma_m) = \exp(i\pi a(\gamma_1) \cdots a(\gamma_m))$$

$\Rightarrow$ multi-linear degree m polynomial (Clifford hierarchy at level m)

Transversality in gauge field formalism

- Some QEC codes support transversal logical gates.
Can we understand it from the viewpoint of the gauge field formalism?
→ (Probably) Yes
- Example: Logical S gate in $[[7,1,3]]$ Steane code (color code)
 - The logical S gate can be implemented by transversal $S^\dagger$ gate

$$\bar{S} = \prod_{i=1}^7 S_i^\dagger = \prod_{i=1}^7 S_i \cdot \prod_{i=1}^7 Z_i$$

- On the other hand, our results give

$$\bar{S} = \prod_{i=1}^7 S_i \cdot \prod_{i < j} CZ_{i,j}$$

⇒ How can we understand this?

Transversality in gauge field formalism

- We can show that the product of the CZ layers and the transversal Z gates is a logical identity operator:

$$\prod_{i < j} CZ_{i,j} \cdot \prod_{i=1}^7 Z_i \sim I$$

⇒ Our result is consistent with the transversal logical S gate:

$$\prod_{i=1}^7 S_i \cdot \prod_{i < j} CZ_{i,j} \sim \prod_{i=1}^7 S_i \cdot \prod_{i=1}^7 Z_i = \prod_{i=1}^7 S_i^\dagger$$

- The above relation holds as an identity on the code space, independent of construction of the logical S gate
- Studying logically identity operators in QEC codes may give us a new insight into transversality of logical gates

Transversality in gauge field formalism

- For $\gamma = (1\ 1\ 1\ 1\ 1\ 1\ 1)$,

$$\prod_{i < j} CZ_{i,j} \cdot \prod_{i=1}^7 Z_i = \exp\left(\frac{i\pi}{2}(a(\gamma)^2 + a(\gamma))\right) =: \mathcal{O}$$

- We can show this operator commutes with all the stabilizers and the logical Pauli operators:

$$[\mathcal{O}, S^X(v)] = [\mathcal{O}, \bar{X}] = 0$$

$\Rightarrow \mathcal{O}$ is a logical identity operator ($\mathcal{O} \sim I$)